

LETTER TO THE EDITOR

A new type of table one: showing instead of telling



To the Editor,

Scientific progress relies on studies that test interventions against predefined outcomes, yet unrecognized biases and hidden confounders can undermine their validity. In 1982, *The New England Journal of Medicine* published one of the first best-practice research guidelines, highlighting the importance of randomization, statistical analysis, and the reporting of baseline characteristics [1]. Then, in 2007 and 2010, the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) and Consolidated Standards of Reporting Trials (CONSORT) guidelines further articulated best practices for reporting

on study population using Table 1 [2,3]. However, the current Table 1 design still has limitations.

Traditional Table 1 formats summarize continuous variables using the mean and standard deviation, or the median and interquartile range, and categorical variables using counts and percentages. Although useful, these metrics reduce complex data to single point estimates that can obscure underlying distributions and limit meaningful comparison between study groups. This concern is particularly relevant in modern clinical datasets, where variables are often overdispersed or contain many zero values. In such settings, reporting only the mean and standard deviation can mislead readers, as extreme values can disproportionately affect both measures. At the same time, CONSORT described statistical testing of baseline differences as illogical and potentially misleading, especially in large datasets

Table 1

Study group characteristics presented in text-only form

Covariance comparison				
Variable	Group 1	Group 2	Group 3	Group 4
Group count	2051	2012	2019	2059
Age (median [IQR])	64.4 [49.7–74.3]	63.8 [48.6–73.7]	64.2 [49.3–73.3]	64.7 [49.2–73.9]
Age (mean $\pm$ SD)	60.8 (18.0)	60.5 (17.9)	60.5 (17.7)	60.9 (18.3)
LOS HRS (median [IQR])	152.7 [90.7–278.8]	149.2 [85.9–275.7]	147.9 [86.7–274.9]	148.7 [86.6–280.2]
LOS HRS (mean $\pm$ SD)	246.0 (335.8)	250.2 (355.1)	256.1 (455.2)	245.9 (319.1)
Gender (female)	1000 (48.8%)	950 (47.2%)	997 (49.4%)	1027 (49.9%)
Gender (male)	1051 (51.2%)	1062 (52.8%)	1022 (50.6%)	1032 (50.1%)
Race (American Indian or Alaska Native)	8 (0.4%)	7 (0.3%)	7 (0.3%)	2 (0.1%)
Race (Asian)	59 (2.9%)	41 (2.0%)	61 (3.0%)	64 (3.1%)
Race (Black or African American)	565 (27.5%)	561 (27.9%)	530 (26.3%)	535 (26.0%)
Race (Native Hawaiian or other Pacific Islander)	3 (0.1%)	1 (0.0%)	3 (0.1%)	1 (0.0%)
Race (unknown)	138 (6.7%)	142 (7.1%)	107 (5.3%)	142 (6.9%)
Race (White)	1278 (62.3%)	1260 (62.6%)	1311 (64.9%)	1315 (63.9%)
Diabetes	823 (40.1%)	816 (40.6%)	816 (40.4%)	811 (39.4%)
Hyperlipidemia	1003 (48.9%)	964 (47.9%)	976 (48.3%)	1022 (49.6%)
Hypertension	1138 (55.5%)	1146 (57.0%)	1097 (54.3%)	1166 (56.6%)
Hypothyroid	445 (21.7%)	457 (22.7%)	429 (21.2%)	444 (21.6%)
Obesity	301 (14.7%)	283 (14.1%)	293 (14.5%)	319 (15.5%)
Reflux	664 (32.4%)	653 (32.5%)	619 (30.7%)	609 (29.6%)
Vitamin D deficiency	165 (8.0%)	158 (7.9%)	143 (7.1%)	173 (8.4%)

IQR, interquartile range; LOS HRS, hospital length of stay in hours; SD, standard deviation.

A

New Table 1 - Covariance Comparison

Groups	Total Patients	Age Mean (SD)	Age Distribution	Female Male	Race Categories* (R1,R2,R3,R4,R5,R6)	Diabetes	Dizziness	Hyperlipidemia	Hypertension	Hypothyroidism	Obesity	Reflux	Vitamin D Deficiency	LOS Median [IQR]	LOS Distribution
1		60.8 (18)		1000 (48.8%) 1051 (51.2%)		40.1%	5.0%	48.9%	55.5%	21.7%	14.7%	32.4%	8.0%	152.7 [90.7–278.8]	
2	2012	60.5 (17.9)		950 (47.2%) 1062 (52.8%)		40.6%	5.8%	47.9%	57.0%	22.7%	14.1%	32.5%	7.9%	149.2 [85.9–275.7]	
3	2019	60.5 (17.7)		997 (49.4%) 1022 (50.6%)		40.4%	5.4%	48.3%	54.3%	21.2%	14.5%	30.7%	7.1%	147.9 [86.7–274.9]	
4	2059	60.9 (18.3)		1027 (49.9%) 1032 (50.1%)		39.4%	6.2%	49.6%	56.6%	21.6%	15.5%	29.6%	8.4%	148.7 [86.6–280.2]	

B

Metric	Group 1	Group 2	Group 3	Group 4
Total Patients	2051	2012	2019	2059
Age Mean (SD)	60.8 (18)	60.5 (17.9)	60.5 (17.7)	60.9 (18.3)
Age Distribution				
Female Male	1000 (48.8%) 1051 (51.2%)	950 (47.2%) 1062 (52.8%)	997 (49.4%) 1022 (50.6%)	1027 (49.9%) 1032 (50.1%)
Race Categories* (R1,R2,R3,R4,R5,R6)				
Diabetes	40.1%	40.6%	40.4%	39.4%
Dizziness	5.0%	5.8%	5.4%	6.2%
Hyperlipidemia	48.9%	47.9%	48.3%	49.6%
Hypertension	55.5%	57.0%	54.3%	56.6%
Hypothyroidism	21.7%	22.7%	21.2%	21.6%
Obesity	14.7%	14.1%	14.5%	15.5%
Reflux	32.4%	32.5%	30.7%	29.6%
Vitamin D Deficiency	8.0%	7.9%	7.1%	8.4%
LOS Median [IQR]	152.7 [90.7–278.8]	149.2 [85.9–275.7]	147.9 [86.7–274.9]	148.7 [86.6–280.2]
LOS Distribution				

Figure. New Table 1 figures show the same data as old Table 1 but contain graphics in a horizontal (Figure A) and vertical (Figure B) orientation. Race categories*, R1, unknown; R2, American Indian or Alaska Native; R3, Asian; R4, Native Hawaiian or other Pacific Islander; R5, Black; R6, White. IQR, interquartile range; LOS, hospital length of stay; SD, standard deviation.

where clinically trivial differences can become statistically significant.

To solve these issues, we propose a graphical enhancement to traditional Table 1 using R [4] package called gtExtras [5]. Continuous variables should include both numerical descriptions of central tendency and variation, as well as compact graphical displays such as kernel density plots. Categorical variables should be represented using adjacent bar charts, and Boolean variables using percentage bars. Such an approach would better align Table 1 with human-centered design principles and modern image-rendering capabilities, enabling readers to more intuitively assess distributions, identify group similarities, and detect potential imbalances.

As a proof-of-concept, we applied this approach to demographic data from 8141 patients hospitalized for pneumonia at an academic quaternary care hospital in Dallas, Texas, in 2022. Patients were divided into four equal groups using an encounter ID-based electronic medical record randomization method [6]. Compared with the traditional Table 1, the graphically enhanced version more clearly displays important distributional differences. For example, the age variable had a median only 6% greater than its mean. In contrast, the hospital length-of-stay variable had a mean

that was 67% greater than its median, demonstrating that numerical summaries alone may fail to convey the shape and skewness of the data.

The goals of Table 1 are not only to describe the study population but also to identify possible threats to validity, including confounding, selection bias, and measurement error. Graphical representations can support these goals by revealing features such as skewness, multimodality, and dispersion that are difficult to appreciate from numbers alone. Because these enhancements are readily achievable with current statistical software, Table 1 should evolve beyond print-era conventions. When it comes to baseline characteristics, showing may be better than telling.

CRediT authorship contribution statement

Jakub Furmaga: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Samuel McDonald:** Writing – review & editing, Methodology. **Bhaskar Thakur:** Writing – review & editing, Methodology. **Robert Turer:** Writing – review & editing, Methodology.

Declaration of competing interest

There are no competing interests.

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Data availability

No data were used for the research described in the article.

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